

TITAN A16 – STRAIGHT METAL BLUEPRINT

Revision: SM-V2.1 (Tightened / Production-Ready)

Classification: Manufacturing Specification

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Core Dimensional Standard: 0.075 Φ -Tolerance Protocol

Document Role: Primary Blueprint (Single Source of Truth)

I. MATERIAL COMPOSITION

Component Role

Material

Mass %

Functional Description

Primary Anchor

Tantalum (Ta)

Primary

Core mass, stiffness anchor, resonance stability

Structural Scaffold

Titanium (Ti)

Secondary

Primary load-bearing lattice, HCP backbone

Interfacial Layer

Gold (Au)

Interfacial

Conductive interface, passivation, damping

Boundary Pinning

Carbon (C)

Trace

Grain boundary stabilization, crack arrest

Material Specifications

Minimum purity (all elements): $\geq 99.95\%$

Total impurity limit: $< 0.001\%$

Alloy form: Monolithic billet

Construction method: Machined features (no pre-stacked laminates)

Target grain size: $\leq 15 \mu\text{m}$ (preferred refinement; verification required)

FIG M1 – SEM / Microstructure Reference (Placeholder)

Representative micrograph showing Ti scaffold continuity, Ta anchoring regions, Au interfacial distribution, and carbon-pinned grain boundaries.

II. STRUCTURAL ARCHITECTURE

Backbone Geometry

Load-bearing backbone exhibits golden-ratio-derived node spacing

Radial and axial symmetry preserved about central bore axis

Geometry optimized for torsional stability and vibration suppression

Triple Helix Architecture

Three interwoven helical paths

Angular separation: 120°

Function: torsional suppression, anisotropic stress management, chatter reduction

Surface Finish

Final machined surface roughness: Ra 0.8 – 1.6 µm

No directional scoring permitted on functional faces

FIG 1 – Backbone Geometry Schematic (Placeholder)

FIG 2 – Triple Helix Wireframe View (Placeholder)

III. AXIAL STACK / LAYER PROFILE

Layer

Material

Functional Role

Thickness (mm)

Tolerance

L1

Titanium

Structural scaffold

2.00

±0.03

L2

Gold

Conductive / damping interface

2.00

±0.03

L3

Titanium

Neutral axis

2.00

±0.03

L4

Gold

Conductive / damping interface

2.00

±0.03

L5

Titanium

Structural scaffold

2.00

±0.03

Total Height: 10.00 mm ±0.05

FIG 3 – Axial Stack Cross-Section (Placeholder)

IV. CENTRAL BORE / ZERO-LOSS PATH

Feature

Specification

Tolerance

Notes

Bore Diameter

3.0 mm

± 0.03 mm

Concentric with OD

Bore Concentricity

—

± 0.02 mm

Measured vs. OD datum

Optional Insert

Graphite core

H7/g6 or cryogenic shrink-fit

Optional functional enhancement

FIG 4 – Bore Cross-Section Detail (Placeholder)

V. TOP LATTICE NODE COORDINATES

Pattern: 12-Node Hexagram / Trefoil

Radial distance from bore center: 5.0 mm

Pocket diameter: 2.0 mm ± 0.03 mm

Pocket depth: 1.0 mm ± 0.03 mm

Material alternation: Ti / Au

Node

θ (deg)

r (mm)

X (mm)

Y (mm)

Material

1

0

5.00

+5.000

0.000

Ti

2

30

5.00

+4.330

+2.500

Au

3

60

5.00

+2.500

+4.330

Ti

4
90
5.00
0.000
+5.000
Au
5
120
5.00
-2.500
+4.330
Ti
6
150
5.00
-4.330
+2.500
Au
7
180
5.00
-5.000
0.000
Ti
8
210
5.00
-4.330
-2.500
Au
9
240
5.00
-2.500
-4.330
Ti
10
270
5.00
0.000
-5.000
Au
11
300

5.00
+2.500
-4.330
Ti
12
330
5.00
+4.330
-2.500
Au

FIG 5 – Top Face Node Layout (Placeholder)

VI. MACHINING / MANUFACTURING PROTOCOL

Billet Preparation

Vacuum Induction Melting (VIM)

Oxygen-free environment required

Machining Sequence

Rough-turn OD and faces

Machine central bore

Mill axial stack pockets (L1–L5)

Mill top lattice node pockets

Execute triple-helix toolpaths

Final facing pass to surface finish spec

Thermal Processing

Stress relief: 540°C × 2 hours

Atmosphere: Argon

Cooling: controlled, furnace or inert cooldown

Optional Conditioning

Ultrasonic excitation: 360 Hz

Duration: 15 minutes

Purpose: resonance stabilization

FIG 6 – CAM Toolpath Simulation (Placeholder)

VII. ASSEMBLY & POST-PROCESSING

Machine all features prior to optional insert

Install graphite core (if used)

Perform stress-relief cycle

Optional ultrasonic conditioning

Handle with non-marring gloves only

Protect functional faces during storage

FIG 7 – Assembly Sequence Diagram (Placeholder)

VIII. QUALITY CONTROL (QC TRAVELER)

Step

Inspection Type

Acceptance Criteria

Dimensional

Ø25 mm / Height 10 mm ± 0.05

2

Surface Finish

Ra 0.8 – 1.6 μm

3

Concentricity

Bore vs OD $\leq \pm 0.02$ mm

4

Grain Size

≤ 15 μm (SEM / EBSD)

5

Resonance Sweep

Peak near 360 Hz

FIG 8 – QC Reference Illustration (Placeholder)

IX. BILL OF MATERIALS (BOM)

Item

Quantity

Notes

Titan A16 Base Alloy Billet

1

Primary material

Graphite Insert (3 × 10 mm)

1

Optional

Synthetic HP Machining Oil

As required

Heat management

PCD Tooling

As required

Final passes

FIG 9 – BOM / Exploded View (Placeholder)

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